

Question

Teacher Hui Hui obtained a white powder **A** in the laboratory one day. **A** is a binary compound containing the non-metallic element **X**, which can quantitatively absorb carbon monoxide from the air and produce another gas present in the air along with the elemental form of **X**. After dissolving **A** in cold water, an excess of sodium hydroxide solution was added to obtain a certain solution. When an excess of chlorine gas was passed through this solution, a colorless solution of **B** was obtained (Reaction 1), with excess sodium hypochlorite remaining in the solution. A pale pink hydrated crystal **C** (containing the metallic element **Y**) was mixed with the aforementioned colorless solution. After thorough reaction, a bright red solution was obtained, and upon adding anhydrous ethanol, a hydrated crystal **D** precipitated. **D** contains no sulfur or chlorine elements.

To verify **D** (assuming it is a pure chemical composition), Teacher Hui conducted the following validation experiments:

- 1) A 0.5000 g sample of **D** was heated at 150°C until constant weight was achieved, leaving a residue of 0.3664 g.
- 2) Excess SO₂ gas was passed through a solution containing 0.5000 g of **D** (Reaction 2). After thorough reaction, the excess SO₂ was removed by heating, and the resulting solution was divided into two portions. To the first portion, a slight excess of silver nitrate solution was added, yielding 0.1452 g of a yellow precipitate **E**. After centrifuging to remove **E**, a slight excess of barium nitrate solution was added to the filtrate. The resulting precipitate was washed, dried, and weighed to yield 0.6252 g of a white precipitate **F**. The second portion of the solution was thoroughly cooled, and a slight excess of NH_4HS solution was added, producing a pink precipitate **G** containing the element **Y**. **G** is a binary compound. After drying at 100°C under a nitrogen atmosphere, the weight of **G** was measured as 0.0179 g.

Additional information:

- 1) The metallic element **Y** commonly occurs in nature as a binary compound **H**, in which the mass fraction of **Y** is 63.2%.
- 2) The aqueous solution of **C** can react with barium chloride solution to also produce the white precipitate **F**.

3) The mass fraction of element **X** in **B** is 43.18%, and the mass fraction of element **Y** in **C** is 19.84%.

Please deduce what **A, B, C, E, F, G, H, X, Y** are. Based on the answers, first calculate and deduce the chemical formula of **D**, then deduce the chemical equations for Reactions 1 and 2 (not ionic equations). Select the appropriate options based on the results.

A. The total number of atoms in the chemical formula of **A** is 3, the total number of atoms in the chemical formula of **B** is 10, and there are 5 mol of water of crystallization in 1 mol of **C**.

B. The total number of atoms in chemical formula **A** is 4, the total number of atoms in chemical formula **B** is 11, and there are 6 mol of crystal water in 1 mol of **C**.

C. The total number of atoms in chemical formula **A** is 5, the total number of atoms in chemical formula **B** is 12, and there are 7 mol of crystal water in 1 mol of **C**.

D. The total number of atoms in chemical formula **A** is 6, the total number of atoms in chemical formula **B** is 13, and there are 8 mol of crystal water in 1 mol of **C**.

E. **A** The total number of atoms in the chemical formula is 7.
B The total number of atoms in the chemical formula is 14.
1 mol **C** contains 9 mol of crystal water.

F. **D** is a relatively complex compound with a 1:1 molar ratio of **X** to **Y**, containing fewer than 10 molecules of water of crystallization; the total number of metal element atoms in the chemical formula of **D** is 5.

G. **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 2:1, containing fewer than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 6.

- H.** **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 1:2, along with fewer than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 7.
- I.** **D** is a relatively complex compound, with a molar ratio of **X** to **Y** of 3:1, and containing fewer than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 9.
- J.** **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 1:3, containing fewer than 10 water molecules of crystallization. The total number of metal element atoms in the chemical formula of **D** is 10.
- K.** **D** is a relatively complex compound in which the molar ratio of **X** to **Y** is 1:1, with more than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 5.
- L.** **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 2:1, containing more than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 6.
- M.** **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 1:2, containing more than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 7.
- N.** **D** is a relatively complex compound in which the molar ratio of **X** to **Y** is 2:2, with more than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 8.
- O.** **D** is a relatively complex compound, with a molar ratio of **X** to **Y** of 3:1, and more than 10 water molecules of crystallization; the total number of metal element atoms in the chemical formula of **D** is 9.

- P.** **D** is a relatively complex compound with a molar ratio of **X** to **Y** of 1:3, containing more than 10 molecules of water of crystallization; the total number of metal element atoms in the chemical formula of **D** is 10.
- Q.** The product of the sum of coefficients on the left side and the sum of coefficients on the right side of Reaction 1 is 32, while the sum of the coefficients on the left side and the right side of Reaction 2 is 35.
- R.** The product of the sum of the coefficients on the left side and the sum of the coefficients on the right side of Reaction 1 is 35, and the sum of the coefficients on the left side and the right side of Reaction 2 is 32.
- S.** The product of the sum of coefficients on the left side and the sum of coefficients on the right side of reaction 1 is 35, while the sum of the coefficients on the left side and the right side of reaction 2 is 38.
- T.** The product of the sum of the coefficients on the left side and the sum of the coefficients on the right side of Reaction 1 is 38, and the sum of the coefficients on the left side and the right side of Reaction 2 is 35.
- U.** The product of the sum of the coefficients on the left side and the sum of the coefficients on the right side in Reaction 1 is 36. The sum of the coefficients on the left side and the right side in Reaction 2 is 40.
- V.** The product of the sum of coefficients on the left side and the sum of coefficients on the right side in Reaction 1 is 40, while the sum of the coefficients on the left side and the right side in Reaction 2 is 36.
- W.** Among the above options, the number of correct options is greater than 3.
- X.** Among the above options, the number of correct options is 2~3
- Y.** None of the above options is correct

Answer

Correct Answer: **S**

Detailed Explanation

First, we must deduce both the elements and compounds:

Based on the description, **A** is a white powder of a binary compound containing nonmetal element **X**, which can quantitatively absorb carbon monoxide and produce another gas in the air along with the elemental form of **X**. We know that the classic white reagent used to quantitatively absorb CO is iodine pentoxide I_2O_5 , meaning **A** is I_2O_5 .

CHECKPOINT

0.5 PTS

A is I_2O_5

Dissolving **A** in cold water produces HIO_3 , which is neutralized by excess $NaOH$ and oxidized under alkaline conditions to form periodate. The relative atomic mass of iodine is 126.90, and based on the mass fraction of 43.18%, the chemical formula of **B** should be $Na_3H_2IO_6$.

CHECKPOINT

1 PTS

The chemical formula of **B** is $Na_3H_2IO_6$

We know that **C** is a pale pink crystalline hydrate containing metal element **Y**. From the pale pink color, we can infer that it is likely a hydrate containing divalent manganese ions Mn^{2+} . The solution of **C** reacts with barium chloride to form a white precipitate, indicating the presence of sulfate ions. Thus, the white precipitate **F** is

$BaSO_4$, and **C** should be $MnSO_4 \cdot nH_2O$. Based on the mass fraction, $n \approx 7$, so we can deduce that **Y** is Mn .

CHECKPOINT

1 PTS

C is $MnSO_4 \cdot 7H_2O$

SO_2 is a strongly reducing gas. Passing it through a solution containing **D** should reduce the high-valent manganese ions and periodate ions. Adding silver nitrate solution to the reduced solution yields a yellow precipitate **E**. Given the properties of iodide and silver ions, **E** is confirmed to be AgI .

CHECKPOINT

0.5 PTS

E is AgI

In another portion of the solution, adding NH_4HS produces a pink precipitate **G** containing Mn . Manganese ions react with sulfide or hydrosulfide ions to form pink manganese sulfide, meaning **G** is MnS .

CHECKPOINT

0.5 PTS

G is MnS

Next, we should analyze how to determine the chemical formula of **D**:

Based on the reactants that produce **D**, **D** should contain high-valent manganese, periodate ions, and water of crystallization. Under the influence of excess SO_2 , the high-valent manganese is reduced to Mn^{2+} , and the periodate ions are reduced to I^- .

In 0.2500 g of **D**:

$$n(I) = n(AgI) = 0.1452/234.8 = 6.184 \times 10^{-4}(\text{mol})$$

CHECKPOINT

0.5 PTS

$$\text{Calculated } n(I) = 6.184 \times 10^{-4}(\text{mol})$$

$$n(Mn) = n(MnS) = 0.0179/87.00 = 2.06 \times 10^{-4}(\text{mol})$$

CHECKPOINT

0.5 PTS

$$\text{Calculated } n(Mn) = 2.06 \times 10^{-4}(\text{mol})$$

Total sulfate ions produced by the oxidation of SO_2 :

$$n(SO_4^{2-}) = n(BaSO_4) = 0.6252/233.4 = 2.679 \times 10^{-3}(\text{mol})$$

CHECKPOINT

0.5 PTS

$$\text{Calculated } n(SO_4^{2-}) = 2.679 \times 10^{-3}(\text{mol})$$

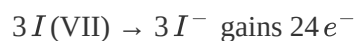
It follows that $n(Mn) : n(I) : n(SO_4^{2-}) = 1:3:13$

CHECKPOINT

1 PTS

Deduced $n(\text{Mn}) : n(\text{I}) : n(\text{SO}_4^{2-}) = 1:3:13$

From the conservation of electrons in the redox reaction:



Thus, $\text{Mn}(\text{X}) \rightarrow \text{Mn}(\text{II}) \sim \text{should gain } 2e^-$

Therefore, the oxidation state of Mn in **D** is +4, i.e., $\text{X}=+4$

CHECKPOINT

2 PTS

Deduced the oxidation state of Mn in **D** is +4

At this point, it is clear that **D** contains 1 Mn , 3 IO_6^{5-} , and several water molecules. Due to the significant number of anions, **D** should also contain several Na^+ or H^+ to balance the charge. (These H^+ may be bound to the periodate ions.)

Assume the chemical formula of **D** is $\text{Na}_x\text{H}_y\text{Mn}(\text{IO}_6)_3 \cdot z\text{H}_2\text{O}$.

Then $M = 0.2500 / (2.06 \times 10^{-4}) = 1213(\text{g/mol})$

CHECKPOINT

1 PTS

Calculated the relative atomic mass of **D** as 1213(g/mol)

$$M(H_2O) = 1213 \times (0.5000 - 0.3664) / 0.5000 = 324.3(\text{g/mol}), \text{ so } z = 18$$

CHECKPOINT

1 PTS

Calculated there should be 18 water molecules of crystallization

From charge conservation, $x + y + 4 = 3 \times 5$ gives $y = 11 - x$.

$$\text{Also, } 22.99x + 54.94 + 3 \times 222.7 + 1.008(11 - x) + 324.3 = 1213.$$

Solving gives $x = 7$, so $y = 4$.

CHECKPOINT

1 PTS

Calculated **D** contains 7 Na^+

CHECKPOINT

1 PTS

Calculated **D** contains 4 H^+

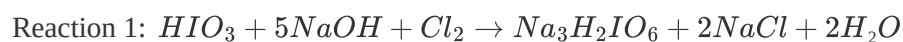
Thus, the chemical formula of **D** is $Na_7H_4Mn(IO_6)_3 \cdot 18H_2O$.

CHECKPOINT

0.5 PTS

Deduced the chemical formula of **D** is $Na_7H_4Mn(IO_6)_3 \cdot 18H_2O$

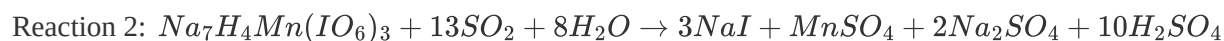
Finally, we can write reactions 1 and 2.



CHECKPOINT

1 PTS

Wrote reaction 1: $HIO_3 + 5NaOH + Cl_2 \rightarrow Na_3H_2IO_6 + 2NaCl + 2H_2O$



CHECKPOINT

1 PTS

Wrote reaction 2: $Na_7H_4Mn(IO_6)_3 + 13SO_2 + 8H_2O \rightarrow 3NaI + MnSO_4 + 2Na_2SO_4 + 10H_2SO_4$

We know that the sum of coefficients on the left side of reaction 1 is 7, and on the right side is 5. For reaction 2, the left side sums to 22, and the right side sums to 16. The product of coefficients for reaction 1 is 35, and for reaction 2, it is 38.